Radiation Theory

A more detailed discussion of radiological theory can be found in the *Troxler Nuclear Gauge Safety Training Program* manual, provided at the Troxler Safety Class.

Atomic Structure

All matter is made up of atoms. For example, water has two atoms of hydrogen (H) and one atom of oxygen (O), which in chemical notation is written H_2O.

An atom is made up of a dense nucleus, consisting of positively charged protons and uncharged neutrons, surrounded by a cloud of negatively charged electrons. Under normal circumstances, the number of electrons in an atom equals the number of protons. The number of protons in the atom is called the *atomic number* (Z). A chemical element consists of all atoms having the same atomic number.

The number of protons plus neutrons in the nucleus is called the *atomic mass* (A). Atoms of a given chemical element can exist in slightly different variants called *isotopes* that have different atomic masses. For example, carbon-12 (C-12) is non-radioactive and carbon-14 (C-14) is radioactive. Isotopes that are radioactive are termed radioisotopes or radionuclides.

Figure 12 depicts a helium atom consisting of two protons and two neutrons in the nucleus and two orbiting electrons.

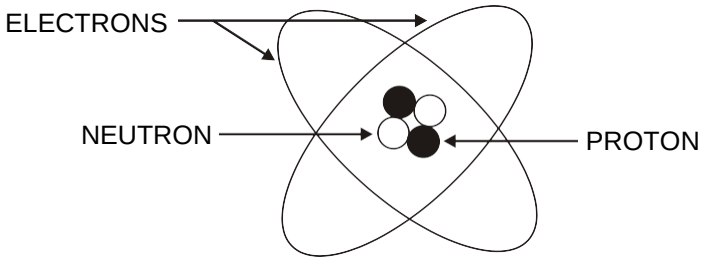


Figure 12. Diagram of an Atom

Radioactivity

Radioactivity is the spontaneous transformation (or disintegration) of an unstable nucleus into a more stable configuration accompanied by the emission of radiation.

The quantity of a radioactive material is measured in terms of the average number of nuclear disintegrations per unit time. The traditional unit of measure for radioactivity (or *activity*) is the *curie* (Ci), which is defined as 3.7×10^{10} disintegrations per second. The activities of the radioactive sources in nuclear gauges are so small that they are typically measured in *millicuries* (mCi), which is one-thousandth of a curie, or *microcuries* (μCi), which is one-millionth of a curie.

In the Standard International (SI) (or *metric*) system, the unit of activity is the *becquerel* (Bq), which equals one disintegration per second. Because the becquerel is such an extremely small unit, the activity of sources in nuclear gauges is normally expressed in *megabecquerel* (MBq), which is one million becquerels, or *gigabecquerel* (GBq), which is one billion Bq.

The radioactivity of a source is not constant, but decreases with time as the source decays. The time it takes for one-half of the original atoms to disintegrate is called the *half-life*. In successive half-lives, the activity decreases to 1/2, 1/4, 1/8 and so on of the initial value. After seven half-lives, less than 1% of the original radioactive atoms remain. Each radioisotope has a characteristic half-life, which can range from seconds to billions of years. The half-lives for the typical radioisotopes used in nuclear gauges are:

Radioisotope	Half-life
Cs-137	30 years
Am-241	432 years

Types of Radiation

The radioactive sources in the gauge produce four types of radiation:

Alpha particles

Beta particles

Gamma rays (photons)

Neutrons

The alpha and beta particles are stopped by the source capsule. Therefore, they present no external hazard to personnel. Only the gamma and neutron radiation from sealed sources contribute to any occupational radiation exposure.

Gamma rays (photons) are a type of electromagnetic radiation, like X rays, radio waves, and visible light. Photons have no mass or electrical charge, and travel at the speed of light. Gamma rays are energetic and penetrating. Dense materials (such as lead, tungsten, and so on) provide the best shielding against gamma radiation.

Neutrons are a form of particulate radiation but, unlike alpha and beta particles, they have no electrical charge. This makes neutron radiation very penetrating. Fast neutrons lose energy primarily by so-called “billiard ball” elastic collisions with the nuclei of low atomic number atoms, especially hydrogen. The best shielding materials for fast neutrons are those with a high hydrogen content, such as water, concrete, and polyethylene.

Radiation Safety

This section discusses the principles of general radiation safety. This information includes specific procedures for operating, inspecting, cleaning, and leak testing the gauge to ensure safe operation.

Radiation Dose

Radiation cannot be detected by any of the human senses (sight, touch, hearing, smell). However, using appropriate instruments and devices, radiation can be detected and measured at levels far below those that significantly affect health.

For purposes of radiation protection, the basic unit of radiation dose is the *rem*. The SI unit is the *sievert* (Sv), where 1 Sv = 100 rem. The rem is a relatively large unit, so often radiation dose is expressed in smaller units called millirem (mrem), where 1 rem = 1000 mrem.

The risk of injury from radiation is generally related to the total radiation dose received over a period of time. It is also related to the dose rate, which is the amount of dose received per unit time. The same amount of radiation received over a long period (months to years) is much less hazardous than if received over a very short period (hours). This has to do with the body's ability to repair cell damage caused by the radiation.

The U.S. Nuclear Regulatory Commission (NRC) has established the following limits on the amount of whole body radiation exposure that individuals may safely receive from licensed radioactive materials.

Type of Individual	Dose Limit
Adult worker	5000 mrem per year
Minor (under 18 years old)	500 mrem per year
Member of the public	100 mrem per year

Limiting Exposure

Under average conditions, an individual working with the gauge will receive less than 200 mrem per year.

A basic principle of radiation protection is that radiation exposure should be kept as far below the limits as is reasonably achievable. This is known as the *ALARA* (as low as reasonable achievable) principle. The three methods for limiting exposure are:

- Time
- Distance
- Shielding

Time

The simplest way to reduce exposure is to minimize the time spent around a radioactive source. If the time spent near a source is cut in half, then the exposure is halved, all other factors remaining constant.

Distance

Distance is another effective means to reduce radiation exposure. A formula known as the *inverse square law* relates the radiation exposure rate to distance (see Figure 13). Doubling the distance from a radiation source reduces the exposure to one-fourth its original value. If the distance is tripled, then the exposure is reduced by a factor of nine, and so on.

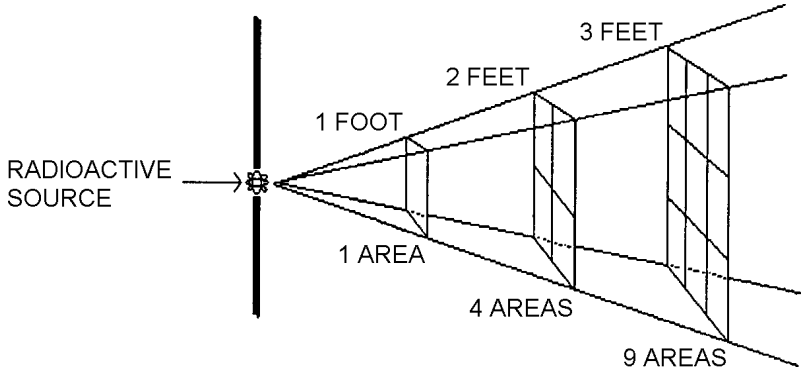


Figure 13. Effect of Distance on Exposure

Shielding

Shielding is any material used to reduce the radiation exposure rate from a radioactive source. The gauge has some built-in shielding, which reduces the exposure rate. When gauges are in storage, additional shielding may be necessary to keep exposure to personnel in adjacent areas below the dose limits for members of the public.

The thickness of any material that reduces the incident radiation intensity by one-half is known as the *half-value layer (HVL)*. The HVL of a material varies with the type and energy of radiation. The HVL values of certain common shielding materials are shown below for gamma and fast neutron radiation.

Material	Cs-137 Gamma Half-Value Layer	Am-241:Be Neutrons Half-Value Layer
Concrete	1.9 in.	4.3 in.
Lead	0.3 in.	*

* Lead does not provide any effective shielding of fast neutrons.

Personnel Monitoring

In the United States, anyone working with or near radioactive materials is subject to occupational dose limits. Individual monitoring of each authorized user is recommended in order to demonstrate compliance with these dose limits.

The most common types of individual monitoring devices used by licensees are film badges and thermoluminescent dosimeter (TLD) badges. Film badges are typically exchanged and processed monthly due to concerns about film fading. TLD badges are usually exchanged quarterly. Troxler offers NVLAP-certified personnel monitoring services using TLD badges.

In Canada, nuclear gauge users are not normally classified as Atomic Radiation Workers. In such cases, the general public dose limit of 500 mrem/year applies. Users may not be required to wear a dosimeter. To establish the personnel monitoring requirements for your application, consult the conditions of your radioisotope license and the CNSC regulatory document *R91: Monitoring and Dose Recording for the Individual*.

Source Encapsulation

The neutron (americium-241:beryllium in the Model 3430 and photon (cesium-137) source materials are welded inside stainless steel capsules. These sealed sources meet U.S. and international regulatory requirements for classification as "Special Form" radioactive material for purposes of transportation. The sealed sources are designed to prevent leakage of radioactive material under severe accident conditions. They are also designed to comply with applicable ANSI classification requirements for sealed sources used in portable gauges.

Source Rod Inspection

To ensure the integrity of the source rod, Troxler recommends that a qualified Troxler service person inspect the gauge and the source rod at least once every five years. This inspection includes checking for excessive wear, corrosion, or damage that could affect the safety of gauge operation.

However, as a precaution if the gauge is ever damaged or dropped, ensure that the tip of the source rod is intact and undamaged (that is, ensure that the source is not missing) as described on page A-2 – A-3.

Contact the Troxler Radiation Safety Department for further advice.

Daily Inspection

The gauge should be inspected daily before use to ensure proper operation of all safety features as follows:

- The source rod opening in the bottom of the gauge is equipped with a spring-loaded tungsten sliding block that shuts when the source rod is in the **SAFE** (shielded) position. To check the operation of the sliding block, push the source rod down into the backscatter position (see Figure on page 3–4), and then raise it back to the **SAFE** position. You should hear a *click* as the sliding block snaps shut. Turn the gauge over and verify that the sliding block is completely shut. If any portion of the opening is uncovered, the sliding block should be cleaned before using, transporting, or storing the gauge. Refer to page D–10 for instructions on cleaning the tungsten sliding block.

⚠ WARNING

Do not store or transport the gauge unless the sliding block is closed. Increased radiation levels may cause excessive personnel radiation exposure and may violate transportation regulations.

If a radiation survey instrument is available, verify that the radioactive gamma source is in place by measuring the exposure rate at the surface of the gauge. The exposure rate should be approximately 10 – 20 mrem per hour. A reading of about 1 mrem or less indicates either that the survey instrument is not working properly or that the cesium-137 source may be missing. Refer to the *Troubleshooting* section of Appendix A for further instructions.

Cleaning the Tungsten Sliding Block

If the tungsten sliding block is not kept clean, it may stick partially or completely open when the source rod is raised to the **SAFE** (shielded) position. This will result in high radiation levels near or in line with the source rod opening on the bottom of the gauge. After cleaning and reassembling the gauge as described below, check the operation of the sliding block by pushing the source rod into the backscatter position, then returning it to the **SAFE** position. You should hear a *click* as the sliding block snaps shut. Inspect the opening on the base of the gauge to confirm that the sliding block is closed. If not, check that the sliding block spring was properly installed after cleaning. If the sliding block still does not close properly, immediately contact the nearest Troxler Service Center. For a list of Troxler and authorized Troxler service centers, refer to the front of this manual or visit the Troxler website at www.troxlerlabs.com/services.

⚠ WARNING

Do not store or transport the gauge unless the sliding block is closed. Increased radiation levels may violate transportation regulations and may cause excessive personnel radiation exposure.

The tungsten sliding block may require cleaning if the source rod becomes difficult to lower into the “measure” position, or if a *click* is not heard when the source rod is raised to the **SAFE** (shielded) position. An improperly operating sliding block may

also result in erratic or incorrect density readings and increased radiation levels.

> **⚠ WARNING**
>
> **Removal of the sliding block results in dose rates of up to one rem per hour in the path of the beam. Stand clear of the gauge bottom while performing this procedure and proceed as quickly as possible while working in the cavity to minimize exposure to your extremities.**

9. With the source rod in the **SAFE** (shielded) position, place the gauge on its side.
10. Clean the heads of the four corner screws that hold the bottom plate to the gauge base (see Figure 14). Using a Phillips screwdriver, remove the four screws in the corner of the plate and remove the plate.
11. To reduce radiological exposure, stand to one side of the gauge. Paying close attention to the position of the sliding block, remove the block.
12. Using a stiff brush or rag soaked in alcohol, clean the sliding block and the cavity.
13. Re-install the sliding block with the angled side up. Apply a light coating of Magnalube-G paste to the **top angled** surface of the sliding block.
14. Re-install the bottom plate. Do not over-tighten screws! Ensure that the source rod moves up and down freely.

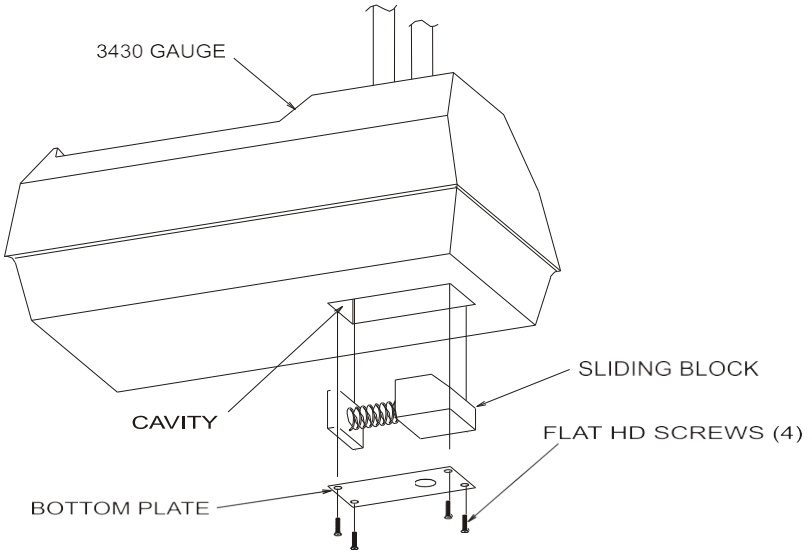


Figure 14. Cleaning the Tungsten Sliding Block

Leak Testing

Unless specified otherwise by your license or state regulations, the gauge must be leak tested at intervals not exceeding 12 months to ensure the integrity of the radioactive source encapsulation. Sample analysis must be performed by a licensed laboratory only.

Using the Troxler Model 3880 Leak Test Kit (PN 102868) and accompanying instructions, perform the following procedure:

⚠ WARNING

Ensure that the source rod is in the SAFE position (see Figure 7 on page 3–4).

1. Write the date, gauge model number, and serial number on the sample form and label.

2. Remove the control panel from the gauge topshell. Locate the yellow radiation label on the top surface of the base.
3. Holding the wipe disk with the tongs, wipe the radiation label.
4. Turn the gauge on its side and locate the opening where the source rod extends through the gauge base.
5. Holding the wipe disk with tongs, wipe the area around and inside the opening where the source rod extends from the gauge base.
6. Pack the disk, as instructed, in the envelope and mail to Troxler Electronic Laboratories, Inc. for analysis.
7. Secure the gauge properly.

Regulatory Requirements

This section summarizes the licensing and training requirements that pertain to ownership or operation of a nuclear gauge. This section also provides information on the proper disposal of the gauge, as well as emergency procedures to follow if the gauge is lost, stolen, or damaged.

Licensing

In the United States, possession and use of the radioactive materials in a nuclear gauge require a license issued by the U.S. Nuclear Regulatory Commission (NRC) or an Agreement State licensing agency. Detailed information on obtaining a license is contained in the *Troxler Licensing Guide*. Copies of this guide are available from Troxler, or can be downloaded from the Troxler website, www.troxlerlabs.com.

To purchase a nuclear gauge in Canada, an owner must obtain a radioisotope license from the Canadian Nuclear Safety Commission (CNSC). The owner should obtain copies of the CNSC Regulations and the Transportation of Dangerous Goods Act and Regulations. For other countries, please consult your local regulatory agency.

Training

In the United States, anyone working with or near radioactive materials must complete a radiation safety training course to be designated an *authorized user*. Authorized users must be trained in the precautions and procedures to minimize radiation exposure; applicable regulatory requirements; and the operating, emergency, maintenance, and transportation procedures for the gauge. Troxler offers training classes designed to meet regulatory agency training requirements for nuclear gauge users.

Disposal

A nuclear gauge contains licensed radioactive material. At the end of a gauge's service life, it must not be discarded as ordinary trash, recycled as scrap material, or abandoned. Instead, a nuclear gauge must be transferred to an authorized recipient licensed by the NRC or an Agreement State. For further information on gauge disposal, contact Troxler.

Emergency Procedures

If the nuclear gauge is lost or stolen, then immediately notify the gauge owner's Radiation Safety Officer (RSO).

The gauge owner should complete the emergency contact information on the lines furnished below. (Note that company refers to the gauge owner's company, not Troxler Electronic Laboratories.) This information should be readily available to the gauge operator at all times.

The company RSO is _____

Call the RSO at _____

The regulatory agency is _____

Call the agency at _____

If a gauge is damaged, then follow the steps below:

1. Locate the gauge and/or sources.
2. Do not touch or move the gauge.
3. Immediately rope off an area around the nuclear gauge and/or sources. A radius of 15 ft. (5 m) is sufficient. Do not leave the area unattended.
4. Keep all unauthorized personnel from the nuclear gauge.
5. If a vehicle is involved, it must be stopped until the extent of contamination, if any, can be established.

6. The gauge user should perform a visual inspection of the nuclear gauge to determine if the source housing or shielding has been damaged.
7. Use a survey meter (such as the TroxAlert Survey Meter) to measure the dose rate at a distance of 1 m (3 ft.) from the gauge.
8. Contact the company RSO (name and number given at the beginning of this section). Provide the RSO with the following:
    a. The date, time, and location of the accident
    b. The gauge model and serial number
    c. The nature of the accident
    d. The location and condition of the gauge and/or source
    e. The dose rate at 1 m (3 ft.) from the gauge
9. If you are unable to reach the RSO, then call your regulatory agency (name and number given at the beginning of this section).
10. Follow the instructions of the RSO. The RSO may need to report the incident to the regulatory agency. The RSO may also be required to notify the USDOT of accidents during transport.
11. Before shipping a damaged gauge to Troxler, obtain an RGA (Returned Goods Authorization) number from the Troxler Customer Service Department, as described in the *Returning the Gauge for Service* section on page C–14.

Gauge Use Precautions

The following precautions should be observed when transporting, storing, maintaining, or operating the gauge.

- Never touch the unshielded tip of the source rod with your bare hands. The dose rates on contact with the source rod tip can result in exposures exceeding the annual dose limits to the skin of the extremities within a short time period (about 1 hour).
- When not taking readings, always keep the source rod in the **SAFE** (shielded) position. For added user safety, the source rod automatically retracts to the **SAFE** position when the gauge is lifted by the handle.
- If you do not hear a *click* when the gauge is raised to the **SAFE** position, look at the bottom of the gauge to verify that the tungsten sliding block is completely closed. If the gauge base opening is not completely closed by the sliding block, the sliding block may require cleaning. Refer to page A–10 for cleaning instructions.
- Do not store or transport the gauge unless the sliding block is completely closed. Increased radiation levels may cause excessive personnel radiation exposure and may violate transportation regulations.
- When preparing a test site, use the drill rod assembly to drill the test hole. Under no circumstances should the source rod of the gauge be used to drill holes.
- Do not tamper with or modify the gauge. Also, do not remove the sealed source from the gauge. Tampering with or modifying the gauge or removing the sealed source can be dangerous. Such actions are illegal unless authorized by your radioactive materials license.

Radiation Profile

Table 3 shows the radiation profile for the Model 3430 gauge. Each table lists the radiation dose equivalent rates (in mrem/hour) for each side of the gauge and transport case shown in Figure 15.

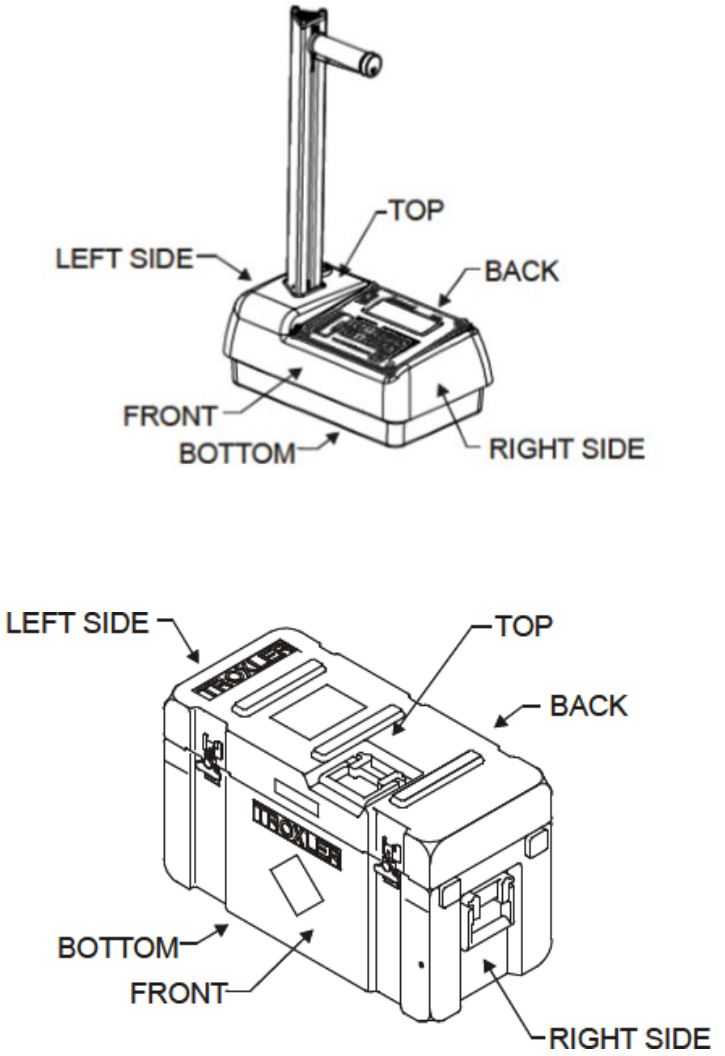


Figure 15. Model 3430 Gauge and Transport Case

Table 3. Radiation Profile for Model 3430 Gauge

(Exposure rate in mrem/hour)

	Surface		5 cm		30 cm		100 cm	
	Gamma	**Neutron**	**Gamma**	**Neutron**	**Gamma**	**Neutron**	**Gamma**	**Neutron**
GAUGE								
Top	30.0	1.0	20.0	1.0	2.0	0.2	§	§
Bottom	19.0	3.0	9.0	2.0	0.6	0.6	§	§
Left Side	25.0	0.6	10.0	0.4	1.2	0.2	0.2	§
Right Side	8.0	1.2	3.0	0.4	1.0	0.2	0.2	§
Front	16.0	1.0	9.0	0.6	1.7	0.2	0.3	§
Back	25.0	1.0	13.0	0.8	2.0	0.3	0.4	§
GAUGE IN TRANSPORT CASE								
Top	17.0	0.2	8.0	0.2	1.2	§	0.2	§
Bottom	6.0	0.2	4.0	0.2	1.0	0.2	0.4	§
Left Side	1.0	§	0.8	§	0.3	§	§	§
Right Side	9.0	2.0	4.0	1.6	0.5	1.0	§	§
Front	9.0	0.4	6.5	0.2	2.0	§	0.2	§
Back	7.0	0.2	5.0	0.2	2.0	§	0.2	§

NOTES:

1. Radiation measurements were of a gauge containing a nominal 8 milli Curies Cesium-137 gamma source and a nominal 40 milli Curies Americium-241: Beryllium neutron source.
2. Gamma measurements were taken with a Bicron MicroRem survey meter, Serial Number B464Y calibrated in January 2011.
3. Neutron measurements were taken with a LudlumModel 12-4 survey meter, Serial Number 140077 calibrated in September 2011.
4. The symbol § denotes a radiation measurement of less than 0.2 milli rem per hour.
5. Measurement position nomenclature for the gauge and transport case is shown in Figures 1 and 2.
6. Orientation of the gauge in the transport case is as follows:
   a. Back of the gauge to the front of the case
   b. Bottom of the gauge to the right side of the case, and
   c. Top of the gauge to the left side of the case.

Appendix E: Standard Count Log

Use the form in this appendix as a guide when recording the daily standard counts. To verify gauge stability, compare the daily standard count to the average of the last four recorded standard counts.

STANDARD COUNT LOG

Gauge Serial Number _____

Date	MS	DS

Date	MS	DS

Appendix F:
Unit Conversion

This chapter contains the following topics and tasks:

- ✓ Measurement units
- ✓ Radiological units

The Model 3430 gauges can display measurement results in either SI (metric) units or English units. Also, HM-181 of 49 CFR changes the standard units of radioactivity in the United States from the English unit of *curies* (*Ci*) to the SI unit of *becquerel* (*Bq*). This requires the shipper to convert the activity on the Bill of Lading from curies to becquerels (GBq). Until everyone is accustomed to the SI units, it is permitted to follow the SI units with the English units in parentheses to clarify the description [for example: 1.48 GBq (40 mCi)].

To help you convert from English units to SI units, the following provides SI conversion factors for common English units relevant to the gauge.

Measurement Units

1 in. = 25.4 mm	1 ft. = 30.48 cm	1 pcf = 16.02 kg/m^3
1 in. = 2.54 cm	1 ft. = 0.3048 m	1 pcf = 1.6 × 10^{-2} g/cm^3

Radiological Units

1 rem = 0.01 Sv

1 Ci = 37 GBq

1 mCi = 37 MBq

1 μCi = 37 kBq

The following table is provided to assist in converting from millicuries to gigabequerels:

mCi to GBq	
8.0	0.30
40	1.48

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